

Product Carbon Footprint Study Report

Product Name: ■■■■

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: ■■■■

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - IPCC 2021

II. Quantitative Purpose

1 kg Ethanol

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg Ethanol.

2. System Boundary

1 kg Ethanol production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: ■■■■

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of ■■■■ produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **5.406 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

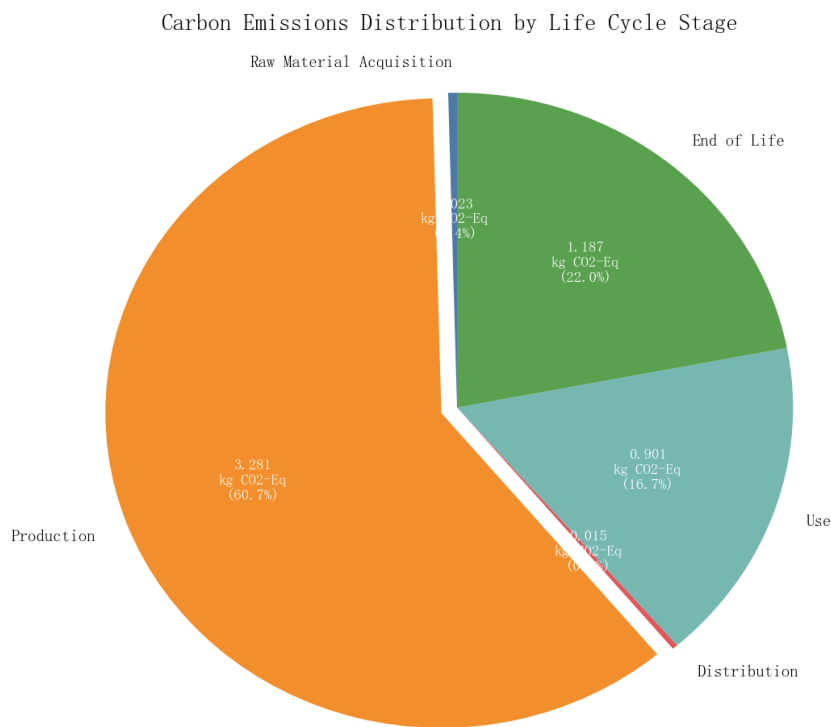


Table 1: Carbon Emissions by Life Cycle Stage

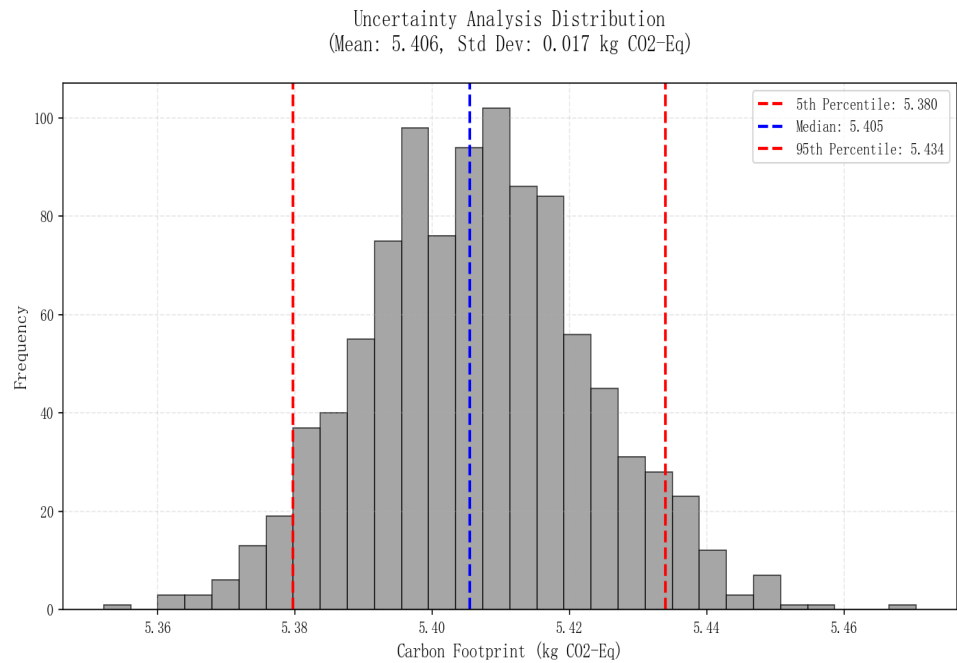
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.023	0.42
Production	3.281	60.68
Distribution	0.015	0.27

Use	0.901	16.66
End of Life	1.187	21.96
Total	5.406	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	5.406
Standard Deviation	0.017
Coefficient of Variation (CV)	0.31%
Median	5.405
5th Percentile	5.380
95th Percentile	5.434

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of ethanol production is calculated at 5.406 kg CO₂-Eq per functional unit. This value is relatively modest compared to many fossil-based commodities, reflecting the biogenic origin of ethanol. However, the uncertainty analysis shows a very low coefficient of variation (0.31%) and a narrow 90% confidence interval (5.380–5.434 kg CO₂-Eq), indicating high precision in the modeled results. The mean value is therefore robust for decision-making purposes.

Analysis of lifecycle stage contributions reveals that the production phase is the dominant emission source, accounting for 60.7% (3.281 kg CO₂-Eq) of the total footprint. This is typical for ethanol manufacturing, where energy-intensive processes such as fermentation, distillation, and drying drive significant fossil CO₂ emissions. The end-of-life stage contributes 22.0% (1.187 kg CO₂-Eq), likely from combustion or degradation of the ethanol. The use stage adds 16.7% (0.901 kg CO₂-Eq), while raw material extraction and distribution are minor, together representing less than 1% of the total.

The major emission sources are concentrated in the production phase, highlighting the need to optimize energy efficiency and fuel switching at the plant level. The relatively high end-of-life contribution also warrants attention, as it may involve direct CO₂ release from biogenic carbon, which is often considered climate-neutral in short-term accounting but still relevant under a full life cycle perspective.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of a generic ethanol production model, likely based on corn or sugarcane feedstock, and the assumption that all biogenic CO₂ released during use and end-of-life is accounted as zero net emissions under conventional carbon neutrality frameworks. The distribution stage is assumed to involve short-distance transport, resulting in a negligible contribution (0.3%). The raw material stage is also assumed to have minimal upstream impacts, possibly due to allocation methods that exclude land-use change or agricultural inputs.

Limitations arise from the data sources: the top process contributions are all listed as "Unknown," indicating that the underlying process-level inventory data are either aggregated, proprietary, or not yet disaggregated. This significantly limits the ability to identify specific emission drivers within the production stage. The coefficient of variation is very low, but this may reflect a simplified model rather than true variability in real-world operations. The quality of the raw material and distribution data is also questionable given their extremely small contributions, which could be artifacts of system boundaries or allocation.

System boundary restrictions include a likely cradle-to-grave scope covering raw material acquisition, production, distribution, use, and end-of-life. However, the exclusion of capital goods, infrastructure, and potential indirect land-use change is not stated. The functional unit is not specified, which limits comparability. Additionally, the "use" stage may only consider direct emissions from ethanol combustion, omitting potential co-product credits or downstream processing.

3. Suggestions for Improvement

- ****Optimize production energy efficiency****: Since production contributes 60.7% of total emissions, prioritize energy-saving measures in distillation and drying units. Implement heat integration, mechanical vapor recompression, or switch to renewable energy sources (e.g., biomass or solar thermal) to reduce fossil fuel consumption at the plant.
- ****Enhance end-of-life carbon management****: With 22.0% of emissions from end-of-life, explore carbon capture and storage (CCS) at ethanol combustion facilities or promote anaerobic digestion of waste ethanol to capture methane for energy recovery, thereby avoiding direct CO₂ release.
- ****Improve use-stage efficiency****: For the 16.7% contribution from the use phase, encourage blending ethanol with gasoline to reduce net emissions per kilometer or promote the use of ethanol in high-efficiency engines (e.g., flex-fuel vehicles with optimized combustion).
- ****Disaggregate process-level data****: To address the "Unknown" process contributions, conduct a detailed process inventory for the production stage. Collect primary data on energy sources, feedstock type, and conversion yields to replace generic assumptions and enable targeted improvements.
- ****Validate distribution and raw material assumptions****: Given the negligible contributions of these stages (0.3% and 0.4%), re-evaluate system boundaries to ensure that transport distances, feedstock cultivation practices, and land-use change impacts are not underestimated. Consider sensitivity analyses to confirm their insignificance.